

广义相对论 第一次作业

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第一章

9. 记发射器的 4-速度为 $\hat{u}_e$ ，观测者的 4-速度为 $\hat{u}_o$ ，光子的 4-动量为 $\hat{p}$ 。

则发射光子的能量为 $E_e = -\hat{p} \cdot \hat{u}_e$ ，观测到光子的能量为 $E_o = -\hat{p} \cdot \hat{u}_o$

于是有：

$$\frac{\lambda_o}{\lambda_e} = \frac{E_e}{E_o} = \frac{-\hat{p} \cdot \hat{u}_e}{-\hat{p} \cdot \hat{u}_o} = \frac{p^0 u_e^0 - \mathbf{p} \cdot \mathbf{u}_e}{p^0 u_o^0 - \mathbf{p} \cdot \mathbf{u}_e}$$

由于发射器在圆心，相对于地面静止， $\mathbf{u}_e = 0$ ，即 $\hat{u}_e = (1, 0, 0, 0)$ ，则有：

$$p^0 u_e^0 - \mathbf{p} \cdot \mathbf{u}_e = p^0$$

又因为圆盘做匀角速度运动，圆盘边缘观测者速度大小为 $|\mathbf{u}_o| = v$ 。速度沿切线方向，与 $\mathbf{p}$ 垂直，则 $\hat{u}_o = \gamma(1, \mathbf{v})$ ，且：

$$p^0 u_o^0 - \mathbf{p} \cdot \mathbf{u}_o = \gamma p^0$$

故：

$$\frac{\lambda_o}{\lambda_e} = \frac{p^0}{\gamma p^0} = \frac{1}{\gamma} = \sqrt{1 - v^2} < 1$$

根据红移因子 z 的定义：

$$z = \frac{\lambda_o - \lambda_e}{\lambda_e} = \sqrt{1 - v^2} - 1 < 0$$

即圆盘边缘观测者接收到原点发出的光子波长变短，被蓝移。

10. 不妨设 4-速度为 $\hat{u}_1$ 的飞船为观测者。在观测者实验室中，观测者自身的 4-速度为 $\hat{u} = (1, 0, 0, 0)$ ，另一艘飞船的相对 4-速度为 $\hat{u}_r = \gamma(1, \mathbf{v})$ 。

于是有：

$$\hat{u}_1 \cdot \hat{u}_2 = \hat{u} \cdot \hat{u}_r = \eta_{00} u^0 u_r^0 = -\gamma$$

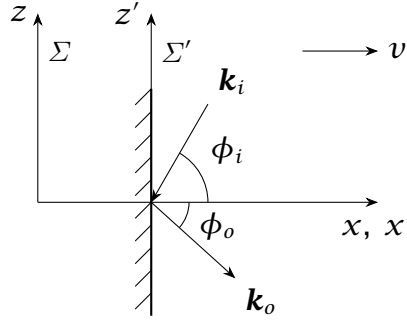
即：

$$(\hat{u}_1 \cdot \hat{u}_2)^2 = (-\gamma)^2 = \frac{1}{1 - v^2}$$

则有：

$$v^2 = \frac{(\hat{u}_1 \cdot \hat{u}_2)^2 - 1}{(\hat{u}_1 \cdot \hat{u}_2)^2}$$

15. 如图，设地面系为 Σ 系，平面镜的静止参考系为 Σ' 系，入射波矢为 $\mathbf{k}_i$ ，反射波矢为 $\mathbf{k}_o$ 。



在 Σ 系中，入射波的 4-波矢为：

$$\hat{k}_i = (\omega_i, \mathbf{k}_i) = (-\omega_i \cos \phi_i, 0, -\omega_i \sin \phi_i, \omega_i)$$

参考系 Σ' 与 Σ 之间的 4-波矢变换为：

$$k'_x = \gamma \left(k_x - \beta \frac{\omega}{c} \right), \quad k'_y = k_y, \quad k'_z = k_z, \quad \frac{\omega'}{c} = \gamma \left(\frac{\omega}{c} - \beta k_x \right)$$

则在 Σ' 系中，入射波的 4-波矢为：

$$\hat{k}'_i = (\omega_i, \mathbf{k}_i) = (-\gamma \omega_i (\beta + \cos \phi_i), 0, -\omega_i \sin \phi_i, \gamma \omega_i (1 + \beta \cos \phi_i))$$

在 Σ' 系中，入射角等于反射角，入射波与反射波频率相等，则反射波的 4-波矢为：

$$\hat{k}'_o = \left(k'_x, k'_y, k'_z, \frac{\omega'_o}{c} \right) = (\gamma \omega_i (\beta + \cos \phi_i), 0, -\omega_i \sin \phi_i, \gamma \omega_i (1 + \beta \cos \phi_i))$$

参考系 Σ 与 Σ' 之间的 4-波矢变换为：

$$k_x = \gamma \left(k'_x + \beta \frac{\omega'}{c} \right), \quad k_y = k'_y, \quad k_z = k'_z, \quad \frac{\omega}{c} = \gamma \left(\frac{\omega'}{c} + \beta k'_x \right)$$

则在 Σ 系中，反射波的 4-波矢为：

$$\hat{k}_o = (\gamma^2 \omega_i (\cos \phi_i + 2\beta + \beta^2 \cos \phi_i), 0, -\omega_i \sin \phi_i, \gamma^2 \omega_i (1 + 2\beta \cos \phi_i + \beta^2))$$

于是有：

$$\omega_o = \gamma^2 \omega_i (1 + 2\beta \cos \phi_i + \beta^2)$$

由 $\gamma = \frac{1}{\sqrt{1 - \beta^2}}$ 得：

$$\omega_o = \omega_i \frac{1 + 2\beta \cos \phi_i + \beta^2}{1 - \beta^2} \quad (1)$$

首先有：

$$\tan \phi_o = \frac{-k_{oz}}{k_{ox}} = \frac{\sin \phi_i}{\gamma^2 (\cos \phi_i + 2\beta + \beta^2 \cos \phi_i)} = \frac{(1 - \beta^2) \sin \phi_i}{\cos \phi_i + 2\beta + \beta^2 \cos \phi_i} \quad (2)$$

或由 $k_o = \frac{\omega_o}{c} \cos \phi_o$ 得：

$$\cos \phi_o = \frac{\cos \phi_i + 2\beta + \beta^2 \cos \phi_i}{1 + 2\beta \cos \phi_i + \beta^2} \quad (3)$$

将 (2)、(3) 两式相乘得：

$$\sin \phi_o = \frac{1 - \beta^2}{1 + 2\beta \cos \phi_i + \beta^2} \sin \phi_i \quad (4)$$

与 (1) 式联立得：

$$\frac{v_o}{v_i} = \frac{\sin \phi_i}{\sin \phi_o}$$

另一方面，注意到 $\tan(\alpha/2) = \frac{\sin \alpha}{1 + \cos \alpha}$ ，则由 (3) 式得：

$$1 + \cos \phi_o = \frac{(1 + \beta)^2 (1 + \cos \phi_i)}{1 + 2\beta \cos \phi_i + \beta^2}$$

与 (4) 式联立得：

$$\frac{\sin \phi_o}{1 + \cos \phi_o} = \frac{(1 - \beta)(1 + \beta)}{(1 + \beta)^2 (1 + \cos \phi_i)} = \left(\frac{1 - \beta}{1 + \beta} \right) \frac{\sin \phi_i}{1 + \cos \phi_i}$$

即：

$$\frac{\tan(\phi_i/2)}{\tan(\phi_o/2)} = \frac{1 + \beta}{1 - \beta} = \frac{c + v}{c - v}$$

第二章

1. 由题意得：

$$\begin{aligned}\varepsilon_{\alpha\beta\gamma\delta}\eta^{\alpha\mu}\eta^{\beta\nu}\eta^{\gamma\rho}\eta^{\delta\lambda}\varepsilon_{\mu\nu\rho\lambda} &= \varepsilon_{\alpha\beta\gamma\delta}\eta^{\alpha\mu}\eta^{\beta\nu}\eta^{\gamma\rho}\varepsilon_{\mu\nu\rho}{}^{\delta} \\ &= \varepsilon_{\alpha\beta\gamma\delta}\eta^{\alpha\mu}\eta^{\beta\nu}\varepsilon_{\mu\nu}{}^{\gamma\delta} \\ &= \varepsilon_{\alpha\beta\gamma\delta}\eta^{\alpha\mu}\varepsilon_{\mu}{}^{\beta\gamma\delta} \\ &= \varepsilon_{\alpha\beta\gamma\delta}\varepsilon^{\alpha\beta\gamma\delta}\end{aligned}$$

最后只需要计算 Levi-Civita 完全反对称张量的完全收缩。

在闵可夫斯基时空下，采用常见的约定 $\varepsilon_{0123} = +1$ 且以度规 $\eta_{\mu\nu} = \text{diag}(-, +, +, +)$ 为准时：

$$\varepsilon_{\alpha\beta\gamma\delta}\varepsilon^{\alpha\beta\gamma\delta} = -1 \times 4! = -24$$

推导如下：

首先通过升指标得到：

$$\varepsilon^{\alpha\beta\gamma\delta} = \eta^{\alpha\alpha'}\eta^{\beta\beta'}\eta^{\gamma\gamma'}\eta^{\delta\delta'}\varepsilon_{\alpha'\beta'\gamma'\delta'}$$

一般规定 $\varepsilon^{0123} = +1$ ，对闵氏度规 $\eta = \text{diag}(-, +, +, +)$ 有：

$$\varepsilon^{0123} = \eta^{00}\eta^{11}\eta^{22}\eta^{33}\varepsilon_{0123} = -1 \times \varepsilon_{0123} = -1$$

$\varepsilon_{\alpha\beta\gamma\delta}$ 在任意有重复索引时为 0，非零项只出现在所有 4! 个对 0、1、2、3 的不同的排列上，而对任意排列 π ，有：

$$\varepsilon_{\pi}\varepsilon^{\pi} = -1 \times (\varepsilon_{\pi})^2 = -1$$

共有 $4! = 24$ 个非零排列项，每项值均为 -1 ，所以求和后为：

$$\varepsilon_{\alpha\beta\gamma\delta}\varepsilon^{\alpha\beta\gamma\delta} = 4! \times (-1) = -24$$

推广地，如果在一个 n 维空间中，度规为对角矩阵，其中有 s 个 -1 和 $n - s$ 个 $+1$ ，则：

$$\varepsilon_{\mu_1\cdots\mu_n}\varepsilon^{\mu_1\cdots\mu_n} = n!(-1)^s$$

2. 令张量 $X^{\mu\nu}$ 的矩阵表示为：

$$X = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 0 & 5 & 3 \\ -2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

V^{μ} 的向量表示为 $V = (0, 3, 1, -2)^{\top}$ ， $\eta_{\mu\nu}$ 的矩阵表示为 $\eta = \text{diag}(-1, 1, 1, 1)$ ， $\eta^{\mu\nu}$ 的矩阵表示为 $\eta^{-1} = \text{diag}(-1, 1, 1, 1)$ 。根据张量和矩阵的计算规则，有：

(a) 缩并降第二个指标相当于右乘 η ，只改变第一列的符号：

$$X^{\mu}_{\nu} = X^{\mu\rho}\eta_{\rho\nu} = X\eta = \begin{bmatrix} -1 & -1 & 0 & 0 \\ 1 & 0 & 5 & 3 \\ 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

(b) 缩并降第一个指标相当于左乘 η ，只改变第一行的符号：

$$X_{\mu}^{\nu} = \eta_{\mu\rho} X^{\rho\nu} = \eta X = \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 5 & 3 \\ -2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

(c) 交换两个指标相当于矩阵的转置：

$$X^{(\mu\nu)} = \frac{1}{2}(X^{\mu\nu} + X^{\nu\mu}) = \frac{1}{2}(X + X^T) = \begin{bmatrix} -1 & -1 & 0 & 0 \\ 1 & 0 & 5 & 3 \\ 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

(d) 缩并升两个指标相当于左乘和右乘 η ：

$$X_{\mu\nu} = \eta_{\mu\rho} X^{\rho\sigma} \eta_{\sigma\nu} = \eta X \eta = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 5 & 3 \\ 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

再计算反对称部分：

$$X_{[\mu\nu]} = \frac{1}{2}(X_{\mu\nu} - X_{\nu\mu}) = \frac{1}{2}[\eta X \eta - (\eta X \eta)^T] = \begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & 1 \\ 1 & -2 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$$

(e) 张量求迹相当于矩阵求迹，但需要先转化为 (1,1)-型张量：

$$X_{\lambda}^{\lambda} = \text{tr}(X_{\nu}^{\nu}) = \text{tr}(X^{\mu\rho} \eta_{\rho\nu}) = \text{tr}(X \eta) = -1 + 0 + 0 + 2 = 1$$

(f) 一阶张量的内积相当于向量的内积，但需要先用度规张量将一个向量降指标：

$$V^{\mu} V_{\mu} = V^{\mu} \eta_{\mu\nu} V^{\nu} = V^T \eta V = (-1) \times 0^2 + 3^2 + 1^2 + (-2)^2 = 14$$

(g) 根据矩阵乘法的定义，要将矩阵 X 的列与 V 的每一个元素相乘再相加，即：

$$V_{\mu} X^{\mu\nu} = W^{\nu}, \quad (W^{\nu})_i = \sum_j V_j X^{ji}$$

对应的矩阵/向量表示为：

$$V_{\nu} X^{\mu\nu} = X^{\mu\nu} \eta_{\nu\rho} V^{\rho} = X^T \eta V = (-5, -1, 15, 5)^T$$

另一种算法是：

$$V_{\nu} X^{\mu\nu} = \eta_{\nu\rho} V^{\rho} X^{\mu\nu} = (\eta V)^T X = (-5, -1, 15, 5)$$

这样得到的结果是一个行向量。

11.(a) 对于 Maxwell 理论, 首先写出电磁场张量 $F_{\mu\nu}$ 的分量形式:

$$F_{\mu\nu} = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{bmatrix}$$

其对应的 (2,0)-型张量 $F^{\mu\nu}$ 的分量形式为: (电场部分符号相反, 磁场部分不变)

$$F^{\mu\nu} = \eta^{\mu\alpha} \eta^{\nu\beta} F_{\alpha\beta} = \begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{bmatrix}$$

这给出: (两矩阵对应元素相乘再求和)

$$F^{\mu\nu} F_{\mu\nu} = -2(E_x^2 + E_y^2 + E_z^2) + 2(B_x^2 + B_y^2 + B_z^2) = 2(B^2 - E^2)$$

$F_{\mu\nu}$ 对应的 (1,1)-型张量 F^μ_ν 的分量形式为: (只改变第一列的符号)

$$F^\mu_\nu = \eta^{\mu\alpha} F_{\alpha\nu} = \begin{bmatrix} 0 & E_x & E_y & E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{bmatrix}$$

而 $F^{\mu\lambda} F^\nu_\lambda$ 是一个二阶张量 $G^{\mu\nu}$, 它的每一个分量满足:

$$G^{ij} = \sum_\lambda F^{i\lambda} F^\lambda_j$$

这相当于矩阵 $F^{\mu\nu}$ 与矩阵 F^μ_ν 转置的乘积:

$$G = F^{\mu\nu} (F^\mu_\nu)^T$$

需要分类讨论如下:

(1) $i = j = 0$ 时:

$$G^{00} = \sum_\lambda F^{0\lambda} F^\lambda_0 = -E_x^2 - E_y^2 - E_z^2 = -E^2$$

(2) $i = 0, j \neq 0$ 时:

$$G^{0j} = \sum_\lambda F^{0\lambda} F^\lambda_j = E_x F^j_1 + E_y F^j_2 + E_z F^j_3 = (\mathbf{E} \times \mathbf{B})_j$$

例如当 $j = 1$ 时:

$$G^{01} = E_x F^1_1 + E_y F^1_2 + E_z F^1_3 = 0 + E_y B_z - E_z B_y = (\mathbf{E} \times \mathbf{B})_x$$

同理:

$$G^{02} = E_z B_x - E_x B_z = (\mathbf{E} \times \mathbf{B})_y, \quad G^{03} = E_x B_y - E_y B_x = (\mathbf{E} \times \mathbf{B})_z$$

因此:

$$G^{0j} = (\mathbf{E} \times \mathbf{B})_j$$

(3) $i \neq 0, j = 0$ 时, 和上一种情况同理:

$$G^{i0} = \sum_{\lambda} F^{i\lambda} F_{\lambda}^0 = F^{i1} E_x + F^{i2} E_y + F^{i3} E_z = -(\mathbf{E} \times \mathbf{B})_i$$

(4) $i \neq 0, j \neq 0$ 时:

$$G^{ij} = \sum_{\lambda} F^{i\lambda} F_{\lambda}^j = F^{i0} F_0^j + F^{ik} F_k^j = -E_i E_j + \sum_k F^{ik} F_k^j$$

其中 $k = 1, 2, 3$ 且 $k \neq i$ 。需要分类讨论如下:

(a) $i = j$ 时:

$$G^{ii} = -E_i^2 + \sum_{k \neq i} F^{ik} F_k^i = -E_i^2 + B^2 - B_i^2$$

例如当 $i = 1$ 时:

$$G^{11} = -E_x^2 + F^{12} F_2^1 + F^{13} F_3^1 = -E_x^2 + B_z^2 + B_y^2 = -E_x^2 + B^2 - B_x^2$$

同理:

$$G^{22} = -E_y^2 + B^2 - B_y^2, \quad G^{33} = -E_z^2 + B^2 - B_z^2$$

因此:

$$G^{ii} = -E_i^2 + B^2 - B_i^2$$

(b) $i \neq j$ 时:

$$G^{ij} = E_i E_j + \sum_k F^{ik} F_k^j = E_i E_j + F^{ik} F_k^j$$

其中 $k = 1, 2, 3$ 且 $k \neq i$ 且 $k \neq j$, 即求和号内 k 只有一个取值。

例如当 $(i, j) = (1, 2)$ 时, $k = 3$, 则有:

$$G^{12} = -E_x E_y + F^{13} F_3^2 = E_x E_y - B_y B_x$$

同理可得:

$$G^{ij} = -E_i E_j - B_i B_j, \quad (k \neq i, k \neq j)$$

则 $i \neq 0, j \neq 0$ 时, G^{ij} 可以统一写为:

$$G^{ij} = \sigma_{ij} = -E_i E_j + (B^2 \delta_{ij} - B_i B_j)$$

也可以用如下方法证明:

$$F^{ik} F_k^j = F^{ik} F^{jk} = (-\varepsilon_{ikm} B^m)(-\varepsilon_{jkn} B^n) = \varepsilon_{ikm} \varepsilon_{jkn} B^m B^n$$

又因为:

$$\varepsilon_{ikm} \varepsilon_{jkn} = \delta_{ij} \delta_{mn} - \delta_{in} \delta_{jm}$$

则有:

$$F^{ik} F_k^j = (\delta_{ij} \delta_{mn} - \delta_{in} \delta_{jm}) B^m B^n = \delta_{ij} (\delta_{mn} B^m B^n) - \delta_{in} B^n (\delta_{jm} B^m) = B^2 \delta_{ij} - B_i B_j$$

综上所述, $G^{\mu\nu}$ 的分量形式为:

$$G^{\mu\nu} = \begin{bmatrix} -E^2 & (\mathbf{E} \times \mathbf{B})_x & (\mathbf{E} \times \mathbf{B})_y & (\mathbf{E} \times \mathbf{B})_z \\ -(\mathbf{E} \times \mathbf{B})_x & \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ -(\mathbf{E} \times \mathbf{B})_y & \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ -(\mathbf{E} \times \mathbf{B})_z & \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}$$

其中:

$$\sigma_{ij} = B^2 \delta_{ij} - (E_i E_j + B_i B_j)$$

于是我们可以写出 $T_{\text{EM}}^{\mu\nu}$ 为:

$$\begin{aligned} T_{\text{EM}}^{\mu\nu} &= F^{\mu\lambda} F_{\lambda}^{\nu} - \frac{1}{4} \eta^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta} \\ &= G^{\mu\nu} - \frac{1}{2} \eta^{\mu\nu} (B^2 - E^2) \\ &= \begin{bmatrix} (E^2 + B^2)/2 & (\mathbf{E} \times \mathbf{B})_x & (\mathbf{E} \times \mathbf{B})_y & (\mathbf{E} \times \mathbf{B})_z \\ -(\mathbf{E} \times \mathbf{B})_x & \sigma_{xx} - (E^2 + B^2)/2 & \sigma_{xy} & \sigma_{xz} \\ -(\mathbf{E} \times \mathbf{B})_y & \sigma_{yx} & \sigma_{yy} - (E^2 + B^2)/2 & \sigma_{yz} \\ -(\mathbf{E} \times \mathbf{B})_z & \sigma_{zx} & \sigma_{zy} & \sigma_{zz} - (E^2 + B^2)/2 \end{bmatrix} \\ &= \begin{bmatrix} (E^2 + B^2)/2 & (\mathbf{E} \times \mathbf{B})_x & (\mathbf{E} \times \mathbf{B})_y & (\mathbf{E} \times \mathbf{B})_z \\ -(\mathbf{E} \times \mathbf{B})_x & \tau_{xx} & \tau_{xy} & \tau_{xz} \\ -(\mathbf{E} \times \mathbf{B})_y & \tau_{yx} & \tau_{yy} & \tau_{yz} \\ -(\mathbf{E} \times \mathbf{B})_z & \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} \end{aligned}$$

其中:

$$\tau_{ij} = \sigma_{ij} - \frac{1}{2} (E^2 + B^2) \delta_{ij} = \frac{1}{2} (E^2 + B^2) \delta_{ij} - (E_i E_j + B_i B_j)$$

对于标量场理论, 注意 $\partial_\mu = (\partial_t, \partial_x, \partial_y, \partial_z) = (\partial_t, \nabla)$, 因此:

$$\eta^{\lambda\sigma} \partial_\lambda \varphi \partial_\sigma \varphi = \eta^{\lambda\lambda} (\partial_\lambda \varphi)^2 = -(\partial_t \varphi)^2 + (\nabla \varphi)^2$$

而:

$$\eta^{\mu\lambda} \eta^{\nu\sigma} \partial_\lambda \varphi \partial_\sigma \varphi = \partial^\mu \varphi \partial^\nu \varphi$$

得到的是一个二阶张量, 每一个分量都是 $\partial^i \varphi \partial^j \varphi$, 即: (注意 $\partial^\mu = (-\partial_t, \nabla)$, 时间分量和 ∂_μ 差一个负号)

$$\begin{bmatrix} (\partial_t \varphi)^2 & -\partial_t \varphi \partial_x \varphi & -\partial_t \varphi \partial_y \varphi & -\partial_t \varphi \partial_z \varphi \\ -\partial_x \varphi \partial_t \varphi & (\partial_x \varphi)^2 & \partial_x \varphi \partial_y \varphi & \partial_x \varphi \partial_z \varphi \\ -\partial_y \varphi \partial_t \varphi & \partial_y \varphi \partial_x \varphi & (\partial_y \varphi)^2 & \partial_y \varphi \partial_z \varphi \\ -\partial_z \varphi \partial_t \varphi & \partial_z \varphi \partial_x \varphi & \partial_z \varphi \partial_y \varphi & (\partial_z \varphi)^2 \end{bmatrix}$$

因此我们有:

$$T_{\text{scalar}}^{\mu\nu} = \partial^\mu \varphi \partial^\nu \varphi - \eta^{\mu\nu} \left[-\frac{1}{2} (\partial_t \varphi)^2 + \frac{1}{2} (\nabla \varphi)^2 + V(\varphi) \right]$$

可以分为三种情况:

(1) $i = j = 0$ 时:

$$T_{\text{scalar}}^{00} = (\partial_t \varphi)^2 - \eta^{00} \left[-\frac{1}{2} (\partial_t \varphi)^2 + \frac{1}{2} (\nabla \varphi)^2 + V(\varphi) \right] = \frac{1}{2} (\partial_t \varphi)^2 + \frac{1}{2} (\nabla \varphi)^2 + V(\varphi)$$

(2) $i = 0, j \neq 0$ 时, $\eta^{0j} = 0$, 则:

$$T_{\text{scalar}}^{0j} = -\partial_t \varphi \partial_j \varphi - \eta^{0j} \left[-\frac{1}{2}(\partial_t \varphi)^2 + \frac{1}{2}(\nabla \varphi)^2 + V(\varphi) \right] = -\partial_t \varphi \partial_j \varphi$$

同理, $j = 0, i \neq 0$ 时:

$$T_{\text{scalar}}^{i0} = -\partial_t \varphi \partial_i \varphi$$

(3) $i \neq 0, j \neq 0$ 时, $\eta^{ij} = \delta_{ij}$, 则:

$$T_{\text{scalar}}^{ij} = \partial_i \varphi \partial_j \varphi - \delta_{ij} \left[-\frac{1}{2}(\partial_t \varphi)^2 + \frac{1}{2}(\nabla \varphi)^2 + V(\varphi) \right]$$

11.(b) 能动张量守恒意味着:

$$\partial_\mu T^{\mu\nu} = 0$$

对于 Maxwell 理论的电磁场张量:

$$\partial_\mu T_{\text{EM}}^{\mu\nu} = \partial_\mu \left(F^{\mu\lambda} F_\lambda^\nu - \frac{1}{4} \eta^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta} \right) = (\partial_\mu F^{\mu\lambda}) F_\lambda^\nu + F^{\mu\lambda} (\partial_\mu F_\lambda^\nu) - \frac{1}{4} \eta^{\mu\nu} \partial_\mu (F^{\alpha\beta} F_{\alpha\beta})$$

在没有外源的情况下, $\partial_\mu F^{\mu\nu} = 4\pi J^\nu = 0$, 即第一项为 0, 因此:

$$\begin{aligned} \partial_\mu T_{\text{EM}}^{\mu\nu} &= F^{\mu\lambda} (\partial_\mu F_\lambda^\nu) - \frac{1}{4} \eta^{\mu\nu} \partial_\mu (F^{\alpha\beta} F_{\alpha\beta}) \\ &= F^{\mu\lambda} (\partial_\mu F_\lambda^\nu) - \frac{1}{4} \partial^\nu (F^{\alpha\beta} F_{\alpha\beta}) \\ &= F^{\mu\lambda} (\partial_\mu F_\lambda^\nu) - \frac{1}{2} F^{\alpha\beta} (\partial^\nu F_{\alpha\beta}) \\ &= F^{\mu\lambda} (\partial_\mu F_\lambda^\nu) - \frac{1}{2} F^{\mu\lambda} (\partial^\nu F_{\mu\lambda}) \end{aligned}$$

现在利用 Bianchi 恒等式:

$$\partial_{[\lambda} F_{\nu\mu]} = \partial_\lambda F_{\nu\mu} + \partial_\nu F_{\mu\lambda} + \partial_\mu F_{\lambda\nu} = 0$$

即:

$$\partial_\nu F_{\mu\lambda} = -\partial_\lambda F_{\nu\mu} - \partial_\mu F_{\lambda\nu}$$

两边同时升 ν 指标得:

$$\partial^\nu F_{\mu\lambda} = -\partial_\lambda F_\mu^\nu - \partial_\mu F_\lambda^\nu$$

于是有:

$$F^{\mu\lambda} (\partial^\nu F_{\mu\lambda}) = -F^{\mu\lambda} (\partial_\lambda F_\mu^\nu) - F^{\mu\lambda} (\partial_\mu F_\lambda^\nu)$$

为了统一偏导数的指标为 μ , 将第一项的 μ 与 λ 交换得: (这不改变正负号)

$$F^{\mu\lambda} (\partial^\nu F_{\mu\lambda}) = -F^{\lambda\mu} (\partial_\mu F_\lambda^\nu) - F^{\mu\lambda} (\partial_\mu F_\lambda^\nu)$$

由于 $F^{\mu\lambda}$ 反对称, $F^{\mu\lambda} = -F^{\lambda\mu}$, 则有:

$$F^{\mu\lambda} (\partial^\nu F_{\mu\lambda}) = F^{\mu\lambda} (\partial_\mu F_\lambda^\nu) - F^{\mu\lambda} (\partial_\mu F_\lambda^\nu)$$

另一方面, F_μ^ν 与 F_μ^ν 也差一个正负号 ($F_\mu^\nu = \eta^{\nu\alpha} F_{\alpha\mu} = -\eta^{\nu\alpha} F_{\mu\alpha} = F_\mu^\nu$), 故:

$$F^{\mu\lambda}(\partial^\nu F_{\mu\lambda}) = 2F^{\mu\lambda}(\partial_\mu F_\lambda^\nu)$$

将其代入 $\partial_\mu T_{\text{EM}}^{\mu\nu}$ 的表达式中得:

$$\partial_\mu T_{\text{EM}}^{\mu\nu} = F^{\mu\lambda}(\partial_\mu F_\lambda^\nu) - \frac{1}{2}F^{\mu\lambda}(\partial^\nu F_{\mu\lambda}) = F^{\mu\lambda}(\partial_\mu F_\lambda^\nu) - \frac{1}{2} \times 2F^{\mu\lambda}(\partial_\mu F_\lambda^\nu) = 0$$

对于标量场的张量:

$$\partial_\mu T_{\text{scalar}}^{\mu\nu} = \partial_\mu \left(\partial^\mu \varphi \partial^\nu \varphi - \eta^{\mu\nu} \left[-\frac{1}{2}(\partial_t \varphi)^2 + \frac{1}{2}(\nabla \varphi)^2 + V(\varphi) \right] \right)$$

或写为:

$$\partial_\mu T_{\text{scalar}}^{\mu\nu} = \partial_\mu \left(\partial^\mu \varphi \partial^\nu \varphi - \eta^{\mu\nu} \left[\frac{1}{2} \partial^\lambda \partial_\lambda \varphi + V(\varphi) \right] \right)$$

第一项:

$$\partial_\mu (\partial^\mu \varphi \partial^\nu \varphi) = (\partial_\mu \partial^\mu \varphi) \partial^\nu \varphi + \partial^\mu \varphi (\partial_\mu \partial^\nu \varphi)$$

第二项:

$$\partial_\mu \left[\eta^{\mu\nu} \left(\frac{1}{2} \partial^\lambda \partial_\lambda \varphi + V(\varphi) \right) \right] = \partial^\nu \left(\frac{1}{2} \partial^\lambda \partial_\lambda \varphi + V(\varphi) \right) = \frac{1}{2} \partial^\nu (\partial^\lambda \partial_\lambda \varphi) + \frac{\partial V}{\partial \varphi} \partial^\nu \varphi$$

由于:

$$\partial^\nu (\partial^\lambda \partial_\lambda \varphi) = \partial^\lambda \varphi (\partial^\nu \partial_\lambda \varphi) + \partial_\lambda \varphi (\partial^\nu \partial^\lambda \varphi) = 2\partial^\mu \varphi (\partial_\mu \partial^\nu \varphi)$$

因此中间两项抵消:

$$\partial_\mu T_{\text{scalar}}^{\mu\nu} = \partial_\mu \partial^\mu \varphi \partial^\nu \varphi - \frac{\partial V}{\partial \varphi} \partial^\nu \varphi = (\square^2 \varphi - \frac{\partial V}{\partial \varphi}) \partial^\nu \varphi$$

根据标量场的运动方程 (Klein-Gordon 方程), $\square^2 \varphi - \frac{\partial V}{\partial \varphi} = 0$, 所以 $\partial_\mu T_{\text{scalar}}^{\mu\nu} = 0$ 。

11.(c) 对于 Maxwell 理论的电磁场张量:

$$T_{\text{EM}}^{00} = \frac{1}{2}(E^2 + B^2)$$

$$T_{\text{EM}}^{ii} = \frac{1}{2}(E^2 + B^2) - (E_i^2 + B_i^2), \quad (i = 1, 2, 3)$$

则张量的迹为:

$$\text{tr}(T_{\text{EM}}^{\mu\nu}) = -T_{\text{EM}}^{00} + T_{\text{EM}}^{11} + T_{\text{EM}}^{22} + T_{\text{EM}}^{33} = -\frac{1}{2}(E^2 + B^2) + 3 \times \frac{1}{2}(B^2 + E^2) - (E^2 + B^2) = 0$$

这说明 (四维) 电磁场的能动量张量无迹。

注意: 对于标量场, $\text{tr}(T_{\text{scalar}}^{\mu\nu}) = -\partial^\lambda \partial_\lambda \varphi - 4V(\varphi)$ 通常是有迹的。